

Bell Inequality Violations from Non-Injective Projection

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Abstract

Violations of Bell inequalities are commonly interpreted as evidence for either nonlocal dynamics or the failure of classical realism. In this work, we show that Bell-type correlations can arise without invoking superluminal influences, retrocausality, or local hidden variables, provided that the mapping between underlying configurations and observable outcomes is non-injective.

We study a class of effective models in which multiple underlying states correspond to the same observable description. Such non-injective projections generically induce non-factorizable joint probability structures at the level of observables, even when the underlying description itself is fully local and deterministic. As a result, the statistical assumptions required to derive Bell inequalities are violated at the effective level, while locality is preserved at the underlying level.

Within this framework, quantum correlations emerge as structural features of projected descriptions rather than as signatures of nonlocal causation. We clarify the precise assumption of Bell's theorem that fails in this setting and show how classical behavior is recovered in regimes where the effective projection becomes approximately injective, suppressing nonlocal correlations through decoherence and environmental coarse-graining.

Our results suggest that Bell inequality violations need not reflect a fundamental nonlocality of nature, but may instead indicate intrinsic information loss induced by non-injective mappings between underlying configurations and observable descriptions.

Keywords: Bell inequalities, quantum correlations, non-injective projection, probabilistic factorization, locality, quantum foundations

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1 Introduction

The experimental violation of Bell inequalities stands as one of the most robust and conceptually challenging features of quantum theory. Repeated tests have confirmed that correlations observed between spacelike separated measurement outcomes cannot be reproduced by any local hidden-variable model satisfying statistical factorization. This empirical fact is commonly interpreted as requiring either nonlocal dynamics, the abandonment of realism, or more radical departures from classical causal intuition.

Despite these conclusions, it is important to recall that Bell inequalities do not constrain physical dynamics directly. They constrain the logical and probabilistic structure of effective descriptions, specifically the assumption that joint outcome probabilities factorize into independent local contributions conditioned on a complete specification of relevant variables. The failure of Bell inequalities therefore indicates a breakdown of effective factorization, but does not, by itself, mandate superluminal influences or causal nonlocality.

In this work, we explore a general and minimal mechanism by which Bell-type correlations may arise without invoking nonlocal dynamics, retrocausality, or hidden variables. The central observation is that effective physical descriptions often arise through many-to-one mappings from underlying configurations to observable outcomes. When this mapping is non-injective, distinct underlying configurations correspond to the same observable description, leading generically to a loss of factorization at the level of observable probabilities.

We study a class of effective models in which observable outcomes are obtained through non-injective projections from an underlying configuration space. The underlying description is assumed to be local and deterministic, but is not fully recoverable from observables. As a result, joint probabilities of outcomes do not admit a factorized representation, even though no superluminal interaction or exchange of information occurs. The violation of Bell inequalities is thus traced to a structural property of the descriptive mapping itself, rather than to any dynamical nonlocality.

Within this framework, quantum correlations are understood as manifestations of global descriptive consistency. Spacelike separated measurements correspond to different local realizations compatible with a single non-factorizable effective description, rather than to independent subsystems endowed with pre-existing local properties. The assumption of outcome determinism conditioned on local variables fails, not because of hidden influences, but because no complete set of outcome-defining variables is accessible at the effective level.

This perspective naturally accounts for the emergence of classical behavior. In regimes where environmental interactions, coarse-graining, or loss of resolution render the effective mapping approximately injective, observable descriptions become effectively factorizable and Bell violations are suppressed. Classical statistical independence thus appears as a limiting case of effective injectivity, rather than as a fundamental principle.

The scope of the present work is deliberately restricted. We do not propose a modification of quantum mechanics, introduce new physical degrees of freedom, or advance a specific dynamical model. Our analysis is purely structural and concerns the logical conditions under which Bell-type inequalities apply to effective descriptions. By isolating non-injective projection as a sufficient mechanism for the failure of factorization, we aim to clarify the conceptual origin of quantum correlations while preserving both operational locality and the empirical content of quantum theory.

2 Non-Injective Projection and the Failure of Bell Factorization

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We consider a minimal abstract setting in which observable descriptions arise from an underlying configuration space through a generally many-to-one mapping. Let Ω denote the space of underlying configurations, assumed to be fully specified and governed by local dynamics. Observable outcomes are defined through a projection

$$\Pi : \Omega \rightarrow \mathcal{O}, \quad (1)$$

where $\mathcal{O}$ is the space of observable states or measurement outcomes. Crucially, Π is assumed to be non-injective, so that multiple distinct elements of Ω correspond to the same observable description.

Let A and B denote two spacelike separated measurement settings, with corresponding observable outcomes $a \in \mathcal{O}_A$ and $b \in \mathcal{O}_B$. In standard local hidden-variable models, one assumes the existence of a variable λ such that the joint probability distribution factorizes as

$$P(a, b \mid A, B, \lambda) = P(a \mid A, \lambda) P(b \mid B, \lambda). \quad (2)$$

Bell inequalities follow from this factorization together with statistical independence assumptions on λ .

In the present framework, no such factorization is generically available. Because the observable outcomes are obtained through the projection Π , the probability distribution

accessible at the effective level is given by

$$P(a, b \mid A, B) = \sum_{\omega \in \Pi^{-1}(a, b)} \mu(\omega), \quad (3)$$

where μ is a probability measure on Ω , and $\Pi^{-1}(a, b)$ denotes the set of underlying configurations compatible with the joint observable outcome (a, b) .

When Π is non-injective, the preimage $\Pi^{-1}(a, b)$ does not decompose into independent subsets associated with A and B . As a result, the joint probability $P(a, b \mid A, B)$ does not admit a representation as a product of marginal probabilities conditioned on any set of effective variables. The failure of factorization arises from the structure of the projection itself, not from any dynamical interaction between the measurement regions.

Importantly, locality at the underlying level is preserved. The measure $\mu(\omega)$ may be defined over configurations with strictly local causal structure, and no superluminal influence or signal exchange is required. The violation of Bell inequalities therefore reflects a mismatch between the factorizable structure assumed in Bell-type derivations and the actual structure of effective descriptions induced by non-injective mappings.

This observation identifies non-injectivity as a sufficient condition for the failure of Bell factorization. Bell inequalities constrain only those effective models in which the mapping from underlying configurations to observables is injective or admits an equivalent factorizable representation. When this condition fails, Bell-type correlations may arise without violating local causality at the underlying level.

4 Bell Inequalities Revisited: Structural vs Dynamical Assumptions

Bell's theorem is commonly presented as a no-go result for local realist theories. More precisely, it establishes that no model satisfying a specific set of probabilistic assumptions can reproduce all quantum correlations. It is therefore essential to distinguish carefully between the assumptions entering the derivation of Bell inequalities and the physical principles they are often taken to represent.

The derivation of Bell inequalities relies on three key assumptions. First, outcome probabilities are assumed to be well-defined and conditioned on a complete specification of relevant variables. Second, joint probabilities are assumed to factorize for spacelike separated measurements, reflecting the absence of direct causal influence between distant measurement settings. Third, the conditioning variables are assumed to be statistically independent of the measurement choices. Violations of Bell inequalities imply that at least one of these assumptions fails.

In many discussions, the failure of Bell inequalities is interpreted as evidence for a breakdown of locality [?]. However, locality in Bell's sense is a condition on the factorization of effective probabilities, not a direct statement about the causal structure of the underlying theory. Bell inequalities constrain models in which observable outcomes can be represented as functions of local settings and a shared set of variables that fully determine those outcomes. They do not directly constrain the existence or absence of superluminal interactions.

Within the framework considered here, the failure of Bell inequalities originates from the breakdown of effective factorization. As shown in the previous section, non-injective mappings between underlying configurations and observable outcomes generically prevent the construction of a complete set of outcome-defining variables at the effective level. Although the underlying description may be local and deterministic, the observable description does not admit a factorized probabilistic representation.

This distinction clarifies the sense in which Bell’s theorem applies. Bell inequalities are valid only for effective descriptions in which observable outcomes can be parameterized by a set of variables that renders the projection from underlying configurations injective or equivalently factorizable. When this condition is not met, the assumptions required for the derivation of Bell inequalities are violated independently of any consideration of causal nonlocality.

From this perspective, violations of Bell inequalities do not compel the introduction of nonlocal dynamics or retrocausal mechanisms. They instead signal a structural limitation of effective descriptions in which observable outcomes are treated as independently specifiable local properties. Nonlocal correlations arise because joint outcomes must be globally consistent with a single non-factorizable effective description, not because information propagates between distant regions.

The present analysis therefore reframes Bell’s theorem as a constraint on a class of descriptive models rather than as a direct statement about the ontology or dynamics of physical reality. Bell inequalities test the compatibility of observed correlations with injective and factorizable effective descriptions. When non-injective projections are involved, Bell-type correlations may arise while preserving locality at the underlying level.

5 Measurement, Decoherence, and the Emergence of Classicality

The structural origin of Bell-type correlations identified above naturally raises the question of their apparent absence in classical macroscopic phenomena. Everyday physical systems exhibit effectively local and statistically independent behavior, despite being ultimately composed of quantum constituents. Any satisfactory account of Bell violations must therefore explain not only their existence, but also their suppression in the classical limit.

Within the present framework, this transition is understood as a change in the effective properties of the projection from underlying configurations to observable descriptions. While the projection is generally non-injective at microscopic scales, environmental interactions, coarse-graining, and limited observational resolution progressively reduce the number of distinct underlying configurations compatible with a given observable outcome. As a result, the effective mapping becomes approximately injective.

Decoherence plays a central role in this process. Through continuous interaction with uncontrolled environmental degrees of freedom, interference between distinct underlying configurations is dynamically suppressed. From the perspective of observable descriptions, this suppression eliminates the global consistency constraints responsible

for non-factorizable joint probabilities. Observable outcomes can then be treated as effectively independent, restoring the factorization required for classical statistical reasoning.

In this regime, joint probabilities admit an approximate decomposition of the form

$$P(a, b \mid A, B) \approx P(a \mid A) P(b \mid B), \quad (4)$$

up to corrections that are exponentially small in the strength of environmental coupling or the degree of coarse-graining. Bell inequalities are therefore not violated in practice, not because the underlying structure has changed, but because the effective description no longer retains access to non-injective correlations.

Importantly, this recovery of classical behavior does not require the introduction of additional collapse postulates or modifications of quantum dynamics. It follows from the same structural mechanism responsible for Bell violations, applied in a regime where the projection becomes effectively one-to-one. Classicality thus corresponds to a limit in which observable descriptions admit an approximately complete specification, rendering joint probabilities effectively factorizable.

From this viewpoint, the quantum-to-classical transition reflects a change in descriptive accessibility rather than a change in fundamental physical laws. Bell-type correlations are suppressed when the information loss associated with non-injective projection is masked by decoherence and coarse-graining. The classical world emerges as a regime in which effective injectivity is restored, and with it, the validity of classical notions of separability and locality.

6 Discussion and Conceptual Positioning

The analysis presented in this work proposes a structural reinterpretation of Bell inequality violations that avoids introducing nonlocal dynamics, hidden variables, or modifications of quantum mechanics. It is therefore important to clarify how this approach relates to existing interpretations and foundational proposals addressing quantum nonlocality.

Unlike local hidden-variable models, the present framework does not assume the existence of a set of accessible variables that fully determine observable outcomes. No completion of the quantum state is postulated, and no additional degrees of freedom are introduced. The failure of Bell inequalities arises not from incomplete knowledge of underlying variables, but from the non-injective nature of the mapping between underlying configurations and observable descriptions.

The present approach is also distinct from superdeterministic models. While superdeterminism denies statistical independence between measurement settings and underlying variables, no such assumption is required here. Measurement choices remain free and uncorrelated with the underlying configuration space. Bell violations arise solely from the structural properties of effective descriptions, not from conspiratorial correlations between settings and states.

Retrocausal interpretations similarly invoke influences propagating backward in time to account for nonlocal correlations. By contrast, the mechanism discussed here is entirely atemporal at the descriptive level. No causal influence, forward or backward,

is required between distant measurement events. Correlations emerge from the global consistency constraints imposed by non-factorizable effective descriptions, rather than from dynamical signaling.

The present framework is compatible with standard accounts of quantum contextuality, but shifts the emphasis from contextual measurement dependence to descriptive non-injectivity. Rather than attributing Bell violations to context-dependent properties of observables, we identify a more general structural origin: the impossibility of assigning independent local outcomes within a single effective description when the underlying mapping is many-to-one.

Finally, we stress that the present work does not advocate a specific ontological interpretation of quantum mechanics. It neither endorses nor rejects realism, operationalism, or informational approaches. Our aim is more modest: to isolate a minimal and sufficient structural condition under which Bell-type inequalities fail. Non-injective projection provides such a condition, clarifying how quantum correlations can arise without abandoning locality at the underlying level.

In this sense, Bell inequality violations may be viewed not as evidence for exotic dynamics, but as indicators of intrinsic limitations of effective descriptions. They reveal the boundaries of applicability of classical probabilistic reasoning, rather than the presence of nonlocal causal mechanisms.

7 Conclusion

We have shown that violations of Bell inequalities can arise as a purely structural feature of effective physical descriptions when the mapping from underlying configurations to observable outcomes is non-injective. In such cases, the probabilistic factorization assumptions required for Bell-type derivations fail at the level of observables, even though locality and causal independence are preserved at the underlying level.

This analysis clarifies that Bell inequalities constrain a class of descriptive models rather than fundamental dynamics. They apply only when observable outcomes admit an injective or factorizable representation in terms of underlying variables. When this condition is not satisfied, Bell-type correlations may emerge without invoking nonlocal interactions, retrocausality, or hidden variables.

We have further shown how classical behavior naturally arises in regimes where environmental interactions and coarse-graining render the effective projection approximately injective. In this limit, joint probabilities recover classical factorization and Bell inequalities are no longer violated in practice.

The present work does not propose a modification of quantum mechanics or a new ontological framework. Its contribution is to isolate a minimal and sufficient structural condition for the failure of Bell inequalities, thereby offering a clarified conceptual perspective on the origin of quantum correlations.

We conclude that Bell inequality violations need not be interpreted as evidence for fundamental nonlocality, but may instead reflect intrinsic limitations of effective descriptions in which observable outcomes are treated as independently specifiable local properties.

Appendices

A Formal Conditions for the Failure of Probabilistic Factorization

This appendix provides a formal clarification of the structural conditions under which probabilistic factorization fails in the presence of non-injective projections. The purpose is not to introduce new physical assumptions, but to make explicit the logical content underlying the arguments presented in the main text.

Let (Ω, Σ, μ) be a probability space describing underlying configurations $\omega \in \Omega$, equipped with a normalized measure μ . Observable outcomes are obtained through a measurable mapping

$$\Pi : \Omega \rightarrow \mathcal{O}, \quad (5)$$

where $\mathcal{O}$ denotes the space of observable descriptions. We assume that Π is non-injective, so that there exist distinct $\omega_1 \neq \omega_2$ such that $\Pi(\omega_1) = \Pi(\omega_2)$.

Let A and B denote two spacelike separated measurement settings, and let $a \in \mathcal{O}_A$ and $b \in \mathcal{O}_B$ denote the corresponding observable outcomes. The joint probability distribution accessible at the effective level is given by

$$P(a, b \mid A, B) = \mu(\Pi^{-1}(a, b)) = \sum_{\omega \in \Pi^{-1}(a, b)} \mu(\omega). \quad (6)$$

Suppose that a factorizable representation of the joint distribution exists. Then there must exist a measurable variable λ and a probability density $\rho(\lambda)$ such that

$$P(a, b \mid A, B) = \int d\lambda \rho(\lambda) P(a \mid A, \lambda) P(b \mid B, \lambda). \quad (7)$$

This representation implicitly assumes that λ provides a complete and non-redundant parameterization of the relevant underlying configurations for the observable outcomes.

However, when Π is non-injective, no such variable λ can generically exist at the effective level. Any candidate λ constructed from observable data must identify entire equivalence classes

$$[\omega] = \{\omega' \in \Omega \mid \Pi(\omega') = \Pi(\omega)\}, \quad (8)$$

rather than individual underlying configurations. As a result, conditioning on λ does not isolate independent contributions associated with A and B .

More formally, if λ is a function of $\Pi(\omega)$, then λ is constant over each equivalence class $[\omega]$. The joint probability conditioned on λ therefore takes the form

$$P(a, b \mid A, B, \lambda) = \sum_{\omega \in [\omega_\lambda]} \mu(\omega \mid \lambda), \quad (9)$$

which does not decompose into a product of independent marginal terms unless additional constraints are imposed. In generic non-injective cases, such constraints are absent, and factorization fails.

This establishes that non-injectivity of the mapping from underlying configurations to observable descriptions is sufficient to obstruct any representation of the form Eq. (??). The failure of Bell-type factorization thus follows from a purely structural property of the descriptive mapping, independently of any assumptions about dynamical nonlocality, retrocausality, or hidden variables.

The result emphasizes that Bell inequalities apply only to effective descriptions admitting an injective or equivalently factorizable representation. When observable outcomes arise through non-injective projections, violations of Bell inequalities are a natural and generic consequence of the descriptive structure itself.

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